

ABOYNE, United Kingdom

WMO: 030800

Lat: 57.0758N

Lon: 2.8411W

Elev: 140

StdP: 99.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-5.1	-7.7	2.0	-5.4	-5.9	2.3	-3.0	14.2	7.7	12.9	7.1	1.7	300	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	23.4	17.3	21.5	16.1	19.9	15.1	18.1	22.0	17.0	20.3	15.9	19.0	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.1	19.4	15.5	11.2	18.5	14.4	10.4	17.5	51.6	22.0	48.1	20.3	45.0	19.0	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.9	9.3	8.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-10.7	26.2	4.8	1.6	-14.2	27.4	-17.0	28.3	-19.7	29.2	-23.2	30.3		
			-10.2	20.3	2.6	2.5	-12.0	22.2	-13.5	23.7	-15.0	25.1	-16.9	26.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	2.8	3.3	4.6	6.6	9.1	11.8	14.1	13.5	11.5	8.3	5.1	3.0	(6)
(7)		DBStd	5.07	3.33	3.39	3.27	3.02	2.92	3.36	2.54	2.54	2.57	3.07	3.33	3.77	(7)
(8)		HDD10.0	1212	223	190	169	109	52	15	1	2	13	69	149	219	(8)
(9)		HDD18.3	3834	482	422	426	352	286	196	134	151	206	309	396	477	(9)
(10)		CDD10.0	423	1	2	2	7	25	69	127	111	58	18	3	1	(10)
(11)		CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0	(11)
(12)		CDH23.3	59	0	0	0	0	3	9	24	22	1	0	0	0	(12)
(13)		CDH26.7	4	0	0	0	0	0	1	1	3	0	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.7	3.8	3.8	3.2	3.1	3.0	2.6	2.6	2.9	3.1	3.2	3.4	(14)
(15)	Precipitation	PrecAvg	915	99	59	56	65	66	66	76	76	70	100	99	84	(15)
(16)		PrecMax	1182	264	118	101	173	134	142	142	166	271	215	216	169	(16)
(17)		PrecMin	611	16	14	9	10	17	22	32	21	22	17	48	38	(17)
(18)		PrecStd	124	52	30	20	42	28	35	31	36	52	48	51	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	14.5	16.4	19.3	22.7	24.5	25.9	26.4	22.5	18.0	15.0	13.7	(19)
(20)		MCWB	10.0	10.7	10.1	12.3	16.2	16.2	19.3	19.2	16.4	14.2	11.7	10.7		(20)
(21)		2%	DB	11.1	12.5	13.5	16.4	19.7	21.4	23.5	22.5	19.9	16.1	13.3	12.2	(21)
(22)		MCWB	8.6	9.2	9.2	11.0	13.8	16.0	17.6	17.2	15.1	12.7	10.7	9.7		(22)
(23)		5%	DB	9.7	10.7	11.8	14.3	17.3	19.4	21.5	20.4	18.1	14.8	11.8	10.6	(23)
(24)		MCWB	7.3	7.9	8.2	10.0	12.3	14.9	16.4	15.7	14.1	12.0	9.4	8.3		(24)
(25)		10%	DB	8.1	9.0	10.4	12.6	15.3	17.8	19.6	18.7	16.6	13.5	10.5	9.0	(25)
(26)		MCWB	6.0	6.4	7.3	8.9	11.0	13.7	15.3	14.6	13.1	11.1	8.5	7.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.3	10.8	13.4	17.1	18.7	20.8	20.5	17.4	14.7	12.4	11.1	(27)
(28)		MDCB	12.4	14.3	14.8	17.9	22.1	22.0	23.8	23.1	21.1	16.9	14.2	13.3		(28)
(29)		2%	WB	8.8	9.4	9.6	11.6	14.5	16.9	18.3	17.6	15.7	13.3	11.0	9.9	(29)
(30)		MDCB	10.9	12.2	13.1	15.4	18.8	19.9	22.1	21.5	18.9	15.4	13.0	12.1		(30)
(31)		5%	WB	7.5	8.1	8.5	10.3	12.8	15.4	17.1	16.4	14.6	12.4	9.7	8.5	(31)
(32)		MDCB	9.4	10.5	11.6	13.8	16.5	18.7	20.4	19.6	17.2	14.3	11.4	10.3		(32)
(33)		10%	WB	6.2	6.7	7.4	9.2	11.4	14.1	15.9	15.4	13.6	11.5	8.8	7.2	(33)
(34)		MDCB	7.9	8.8	10.2	12.3	14.6	17.3	18.8	18.0	16.0	13.2	10.2	8.8		(34)
(35)	Mean Daily Temperature Range	MDBR	6.1	7.1	8.1	9.3	10.3	9.3	9.4	9.6	9.2	7.9	6.9	6.4	(35)	
(36)		5% DB	MCDBR	7.6	8.9	11.0	12.6	14.3	12.9	12.9	12.3	11.9	9.6	7.8	8.1	(36)
(37)		MCWBR	6.3	6.7	7.4	8.0	9.0	8.2	7.8	7.7	7.8	7.1	6.4	6.9	(37)	
(38)		5% WB	MCDBR	7.5	8.0	10.5	11.9	13.2	12.0	11.5	11.2	10.5	8.4	7.5	8.0	(38)
(39)		MCWBR	6.4	6.4	7.4	7.9	8.7	8.3	7.7	7.5	7.5	6.7	6.5	6.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.323	0.351	0.373	0.372	0.370	0.372	0.361	0.345	0.331	0.303	0.285	(40)	
(41)		taud	2.256	2.342	2.319	2.298	2.351	2.380	2.378	2.419	2.472	2.443	2.318	2.198	(41)	
(42)		Ebn at Noon	570	727	801	839	865	870	859	845	807	714	566	471	(42)	
(43)		Edh at Noon	51	71	94	111	112	111	109	98	80	64	47	41	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.46	1.10	2.13	3.48	4.68	4.61	4.46	3.52	2.48	1.32	0.59	0.32	(44)	
(45)		RadStd	0.06	0.10	0.25	0.35	0.26	0.38	0.38	0.23	0.20	0.13	0.06	0.04	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page